To follow along in your workspace, please open the following notebook:

- CH3-01-Creating Embedded Chunks

Our first step for data preparation is to extract the unstructured information from the PDF files. You may have many PDFs, which can be very long. Long documents can potentially be a problem for our future chatbot because they can easily exceed most Large Language Models' maximum context lengths (check out the blog in *Further reading* for more information on context lengths). Additionally, we likely don't need an entire document's worth of text to answer a specific question. Instead, we need a section or "chunk" of this text.

In the context of LLMs, the term "token" refers to a chunk of text that the model reads or generates. A token is typically not a word; it could be a smaller unit, like a character or a part of a word, or a larger one like a whole phrase. For this project, we create chunks of no more than 500 tokens with a chunk overlap of 50 tokens using the `SentenceSplitter` module of the `LlamaIndex` library. You could also use the `LangChain` library or any library you choose to split the content into chunks. We use the open-source `Llama-tokenizer` as this will be our main family of models across our project. Note that tokenizers may play a crucial role in your RAG quality.

If you are unfamiliar with tokenizers or struggling with any of the LLM verbiage, the Databricks LLM learning path (see Further reading) can provide a solid foundation of understanding to build upon. There other sources for learning about LLMs and NLP in general. However, we can only recommend the content of course we know.

Let's go through the steps to create the embedded chunks.

1. To start building the Bronze data layer, create an empty Delta table that includes the table's schema. Use the `GENERATED BY DEFAULT AS IDENTITY` feature to leverage Delta table capabilities to index the newly arriving data automatically. Also, add a table property to set the Change Data Feed to true.

```
%sql
--Note that we need to enable Change Data Feed on the table to create the index
CREATE TABLE IF NOT EXISTS pdf_documentation_text (
  id BIGINT GENERATED BY DEFAULT AS IDENTITY,
  pdf_name STRING,
  content STRING,
  embedding ARRAY <FLOAT>
  ) TBLPROPERTIES (delta.enableChangeDataFeed = true);
```

Figure 3.19 – Create an empty Delta table with the name pdf_documentation_text

2. For the next step, read the raw PDFs from the volume folder and save them to a table named `pdf_raw` (*Figure 3.20*). Store the PDFs in binary format in the `content` column to extract the text later. We will come back to the `pdf_documentation_text` table.